

Report on the Effectiveness of Anti-bullying Strategies in our Schools

For: Tan Mei Ling, Edwin Tan
Lecturers
Nanyang Polytechnic

By: Ashley Wong Jing Xuan
Tia Isabel Solanki
Wong Kah Ann
Ee Rui En Rachel

Admin No.	Student Name	Sub-hypothesis
221067Q	Ee Rui En Rachel	Integrating anti-bullying education into the curriculum will promote empathy, respect, and positive social interactions among youth and thus, mitigate bullying.
220911C	Ashley Wong Jing Xuan	Implementing reporting systems, anonymous reporting options, and responsive procedures for addressing bullying incidents will create a safe and supportive school environment.
230796P	Wong Kah Ann	Providing training and resources for students to become active bystanders and intervene in bullying situations will discourage and prevent bullying incidents.
220892L	Tia Isabel Solanki	Educating students about responsible online behavior, digital citizenship, and the potential consequences of cyberbullying will promote safe and respectful online interactions.

[28/7/2023]

TABLE OF CONTENTS

List of Illustrations	page iii
Abstract	page iv
Introduction	
- Purpose of Report	page 1
- Background Information	page 1
Findings and Discussion	
- Methodology of survey	page 1
- Analysis of Survey Results	pages 2 - 9
Conclusion	page 9
Recommendations	page 9
References	page 10

LIST OF ILLUSTRATIONS

Binomial Distribution Graph 1	page [2]
Pie Chart 1	page [3]
Bar Graph 1	page [3]
Pie Chart 2	page [5]
Linear Regression Graph 1	page [5]
Binomial Distribution Graph 2	page [6]
Pie Chart 3	page [7]
Pie Chart 4	page [7]
Binomial Distribution Graph 3	page [8]
Pie Chart 5	page [8]

ABSTRACT

Bullying among youths, both physical and online, is a serious problem. This report aims to investigate the effectiveness of current anti-bullying strategies in schools. We formulated four sub-hypotheses to gain insights into the main hypothesis. An online survey was conducted to gather data, which was analysed through data analysis methods such as binomial distribution, linear regression, and central tendency. The findings indicate that current strategies are sufficiently effective, though students desire further improvements. Overall, implementing anti-bullying strategies proves effective in managing bullying. We recommend schools prioritize more on educating youths to be civil in their spaces.

INTRODUCTION

Purpose of this Report

The purpose of this report is to examine our hypothesis “Implementing comprehensive anti-bullying strategies in schools will effectively reduce bullying and cyberbullying incidents among youth.”

Background/Current Situation – Preventing Bullying Amongst Students

In 2008, the Singapore Children’s Society conducted a survey of 786 Primary and 513 Secondary School Students and discovered that approximately 1-in-5 Primary Students and 1-in-4 Secondary Students surveyed were victims of bullying.

13 years later in 2021, an article from the Ministry of Education officially commented on the various anti-bullying strategies implemented in our schools and their effectiveness. They found that from 2012 onwards, only 2-in-1000 of Primary Students and 5-in-1000 Secondary Students were victims of bullying.

As observed, the amount of bullying victims among Primary Students had decreased from 20% to an incredible 0.2%. While a similarly impressive trend occurred with the amount of bullying victims among Secondary Students dropping from 25% to 0.5%. Thus, we advocate that implementing anti-bullying strategies in schools will effectively reduce bullying and cyberbullying incidents among youth.

This report seeks to examine the experiences and viewpoints of those who have experienced or witnessed bullying. In order to understand the prevalence and effects of bullying in this surveyed population. It is crucial to note that the survey only provides a glimpse of experiences within the surveyed population that was polled and could not be representative of the general community. For a more thorough knowledge of the occurrence and impacts of bullying, additional study with a bigger and more varied sample would be ideal.

FINDINGS AND DISCUSSION

Methodology

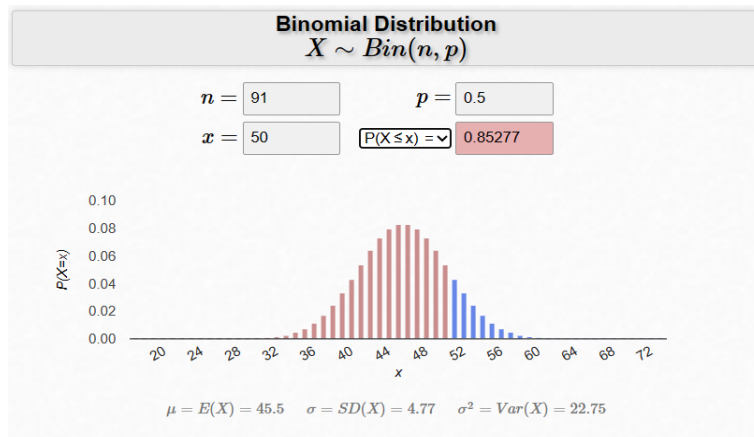
The survey gathered 91 responses through online simple random and convenience sampling, excluding inconclusive neutral responses. The majority of responders (86.8%) were aged 13 to 18 years old, followed by 8.8% aged 19 to 21 years old, and 4.4% aged 22 to 25 years old. In terms of education, the largest group was Polytechnic/NAFA Diploma students (65.9%), followed by Junior College/Millennia Institute students (14.3%),

Secondary School students (6.6%), Institute of Technical Education students (6.6%), and University students (4.4%). Of the remaining 2.2%, 1 was a Primary School student while the other was not receiving schooling.

Analysis of Survey Results

Sub-Hypothesis 1

For this section, we will look into whether **integrating anti-bullying education into the curriculum will promote empathy, respect, and positive social interactions among youth and thus, mitigate bullying.**

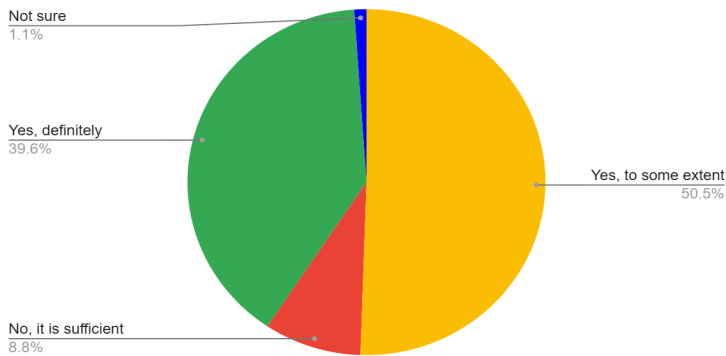


In the survey conducted on the topic of bullying in schools and communities, a total of 91 respondents were asked if they had witnessed or personally experienced bullying. Using Binomial Distribution to find the chance of any student encountering an incident, we made the p-value—the chance of any student witnessing or personally encountering bullying—a 50% split between two mutually exclusive events, 'have' or

'have not'. With the total number of trials, $n = 91$, and successes, $x = 50$, we found that the likelihood of exactly 50 or less out of 91 students witnessing or personally encountering bullying was 85.28%. From the results of Binomial Distribution, the probability of students witnessing or personally experiencing bullying is 0.85277. This indicates that bullying is a prevalent issue affecting a considerable amount of individuals. The findings highlight the importance of providing comprehensive education about the nature, causes, and impacts of bullying, given its profound influence on students' well-being and mental health.

Some comments were provided by students who have encountered bullying. Name-calling, exclusion and cyberbullying were some of the prominent issues, with students being subjected to hurtful insults and humiliation based on their physical appearance, intellectual abilities and racial backgrounds. These findings highlight the importance of comprehensive anti-bullying policies as well as an inclusive and supportive environment in schools and communities.

Do you think there is a need for an increased awareness and education about bullying in your school?



When asked if there was a need for increased awareness and education about bullying in their schools, 90.1% expressed their support for an increased need of the matter.

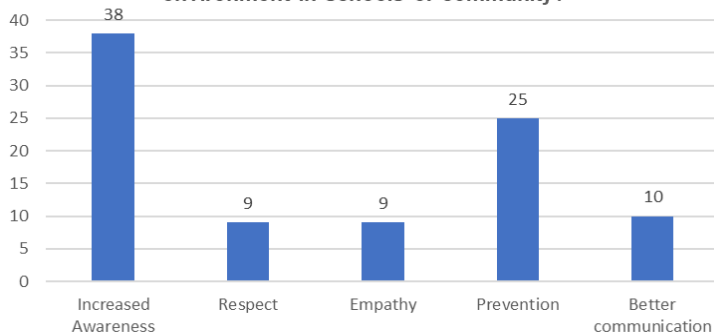
Among these respondents, 39.6% were certain about the need for awareness, while 50.5% agreed to some extent. These findings suggest a clear recognition of the problem's importance and highlight the respondents' awareness of it.

(With Reference to Q1) With cases of mockery and exclusion so prevalent, it points out the necessity for schools to combat a lack of empathy and respect among students and also address discrimination in cliques. Our analysis also discovered instances of exclusion during conversations and the effects of favouritism in the classroom, emphasising the need of promoting positive social connections among youths.

However, 8.8% of respondents expressed that the current anti-bullying strategies and educational programs are already effective in reducing bullying while 1.1% were not sure. Some might believe the current efforts are sufficient due to positive outcomes, while others might be unaware of the severity of bullying, leading them to perceive the current level of awareness as adequate.

In conclusion, while some respondents may believe that current levels of awareness and education are adequate, the majority's perspective emphasises the importance of **continuing efforts to increase awareness and education about bullying** in order to promote empathy, respect, and positive social interactions among youth.

How do you think educating youths about bullying can contribute to creating a safer and more inclusive environment in schools or community?



I have summarised the responses into 5 different categories. The results strongly support the hypothesis. Many students believe that educating youths about bullying builds an inclusive environment through increased awareness, respect, empathy, prevention, and better communication in schools and communities.

In this open-ended question, the majority of respondents, 41.8% (38 out of 91) highlighted that **increased awareness** about bullying can contribute to a safer and more inclusive environment. Respondents said that schools can provide students with the ability to recognize and respond to bullying effectively by raising awareness about different types of bullying, their effects, and preventative techniques. Through education and cultivating awareness, individuals become more mindful of their actions, develop positive habits, and understand that bullying will not be tolerated.

Additionally, 27.5% (25 out of 91) of respondents, underline the importance of proactive efforts to **prevent bullying**. Educating youths can prevent bullying. Respondents mentioned that implementing preventive strategies like anti-bullying policies and awareness campaigns reduces bullying incidents, and fosters a healthy, inclusive school culture that discourages harmful behaviour.

Furthermore, 9.9% (9 out of 91) think that educating youths about bullying results in **increased respect and empathy** towards one another. The responses stressed the importance of respecting differences and being mindful of their words and actions. In addition, education on fostering empathy among youths creates a positive culture where students empathise with victims, develop compassion and consideration for others. Lastly, promoting **better communication with trusted adults** creates a supportive environment where youths are able to freely report and thus address bullying incidents promptly.

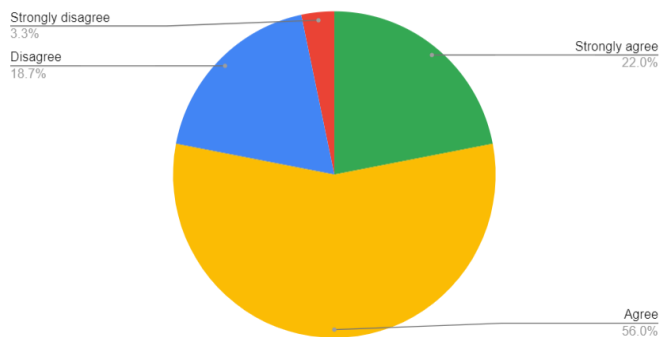
With the prevalence of bullying behaviours evident in the survey, the hypothesis that integrating anti-bullying education into the curriculum can play a critical role in promoting empathy, respect, and positive social interactions among youth gains significance. Such education and increased awareness lead to more pleasant interactions and fewer incidents of bullying and thus a safer environment for all.

Sub-Hypothesis 2

For this section, we will look into whether implementing reporting systems, anonymous reporting options, and responsive procedures for addressing bullying incidents will create a safe and supportive school environment.

We first asked students what policies and procedures their schools implemented. Most students could name their policies and procedures, some included disciplinary consequences such as suspension and detention for the bully. Others mentioned that counselling was provided by school counsellors, teachers and parents. Additionally, students mentioned that their code of conduct ensured a safe and secure learning environment. Educational methods such as disciplinary and cyberbullying awareness talks and posters were also in place.

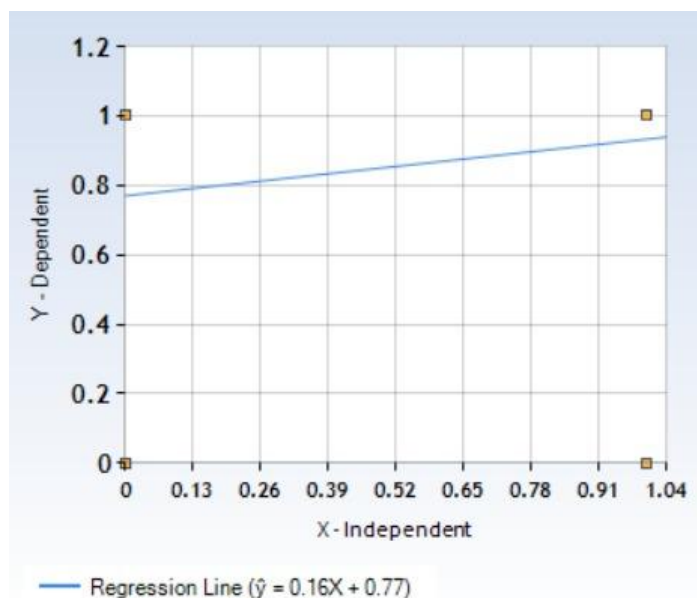
Do you agree that those approaches you listed above are effective?



We also found that 78% of the responders gave a positive response that the approaches in place are effective as shown from the pie chart.

Another question was about the ways schools can improve their current ways of addressing bullying incidents. 65.8% of the respondents think that schools can improve on their current ways by putting more emphasis on education, while 34.2% called for stricter discipline.

We also asked students how familiar they were with the current reporting systems and procedures for addressing bullying incidents in their schools along with how safe they felt within their school environment. Of 91, 46 were unfamiliar while 75 felt safe.



To prepare the data for linear regression analysis, 0 was assigned to negative responses from students indicating “Unfamiliar” and “Unsafe”. Consequently, we organised the responses so that 1 was assigned to positive responses from students indicating “Familiar” and “Safe”. The resulting positive slope in the linear regression graph indicates an overall positive relationship between the reporting systems for bullying incidents and feeling safe at school.

With there being a positive correlation between feelings of safety in the school environment and student familiarity of systems, it shows that as systems are more effective in mitigating bullying, more people come to know them as a result of their sense of security.

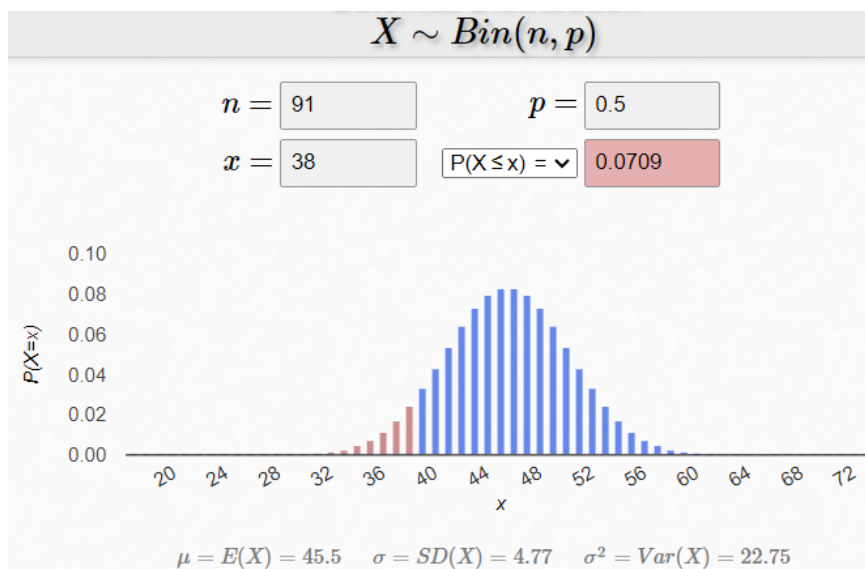
In conclusion, the current policies and procedures in place are effective in creating a safe and supportive school environment, even if more than half of the respondents are unfamiliar with them. However, more

believe that schools should focus on education rather than discipline to staunch cases. Hence, schools should strengthen school policies and procedures before embarking on education.

Sub-Hypothesis 3

For this section, we will examine whether providing training and resources for students to become active bystanders and intervene in bullying situations will discourage and prevent bullying incidents.

We asked 91 students if they had received any training or resources, such as workshops or lessons, on how to be active bystanders in bullying situations. Out of the total responses, only 38 students (41.76%) indicated that they had received such resources.



Using binomial distribution to determine the likelihood of any student getting these resources, we made the p-value—the chance of any student receiving these resources—a 50% split between two mutually exclusive events, 'have' or 'have not'. With the total number of trials, n , as 91, and successes, x , as 38, we found that the likelihood of exactly 38 or less out of 91 students receiving these resources was only 7.09%. In fact, the probability of there being more than 38 of

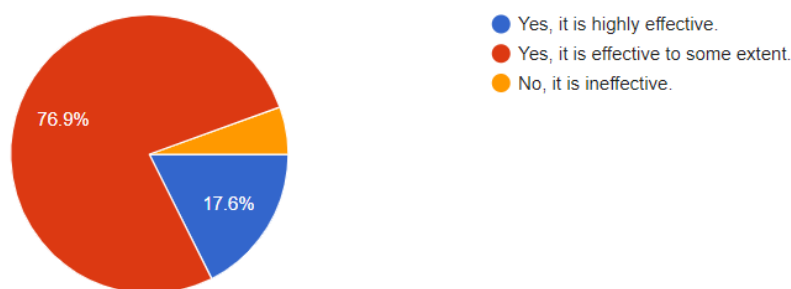
91 students receiving these resources is 92.91%. This proves that there are extremely fewer students receiving training and resources to discourage and prevent bullying incidents than there should be, and we believe it will negatively impact their responses to real incidents.

This inference is supported by our next question, “Have you ever witnessed a bullying incident online or in person where you were able to step in as an active bystander?” Of 55 reported cases from 91 respondents, just 22 managed to intervene in a situation and mitigate the situation, whereas 33 either abandoned or worsened it when they stepped in to mediate. This suggests that when students do not receive proper education and resources to combat bullying they are 1.5 times more likely to exacerbate the situation.

To know why students are 1.5 times more likely to worsen a situation, we asked respondents to rate how confident they would be when faced with an incident on a scale of 1 to 5. In ascending order, 12 (13.2%) were

“Not confident at all”, 23 (25.3%) were “Somewhat confident”, 38 (41.8%) were “Moderately confident”, 6 (6.6%) were “Very confident”, and 12 (13.2%) were “Extremely confident”. This explains that when not enough students are taught to handle bullying cases, the majority find themselves less confident in intervening.

So far, we can see that when not enough students receive resources on how to manage bullying situations, they will likely worsen one. However, to reach a consensus, we asked our respondents if they believe providing training and resources for students to become active bystanders is an effective strategy to discourage and prevent bullying incidents.



Using mode as the measure of central tendency, 70 (76.9%) of 91 students tend to agree that being provided education and resources alone would already prove effective.

Thus, we conclude that providing training and resources for students to become active bystanders and intervene in bullying situations will discourage and prevent bullying incidents because if they are not provided, a student populace would hesitate to step.

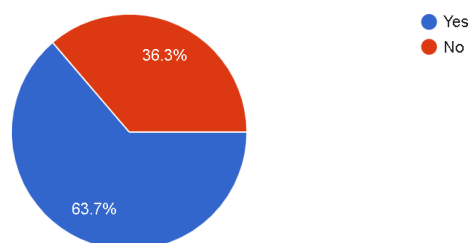
Sub-Hypothesis 4

For this section, we will look into whether educating students about responsible online behavior, digital citizenship, and the potential consequences of cyberbullying will promote safe and respectful online interactions.

First, we asked the students if they have ever been educated or received training in school on either responsible online behavior, digital citizenship, or cyberbullying prevention. Based on our survey results, the majority (58) of the respondents have received training or education on responsible online behavior, digital citizenship, or cyberbullying prevention.

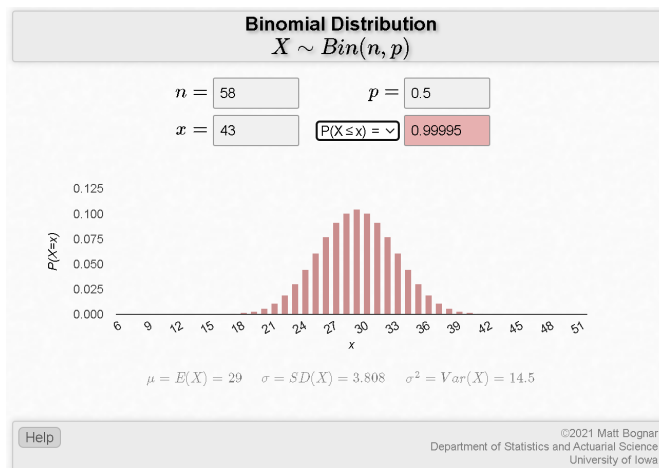
Have you ever been educated or received training in school on either responsible online behavior, digital citizenship, or cyberbullying prevention?

91 responses



The respondents who have received education/training about bullying were then asked if they had noticed a change in their own online behavior after being educated with 74.1% reporting an improvement in their behaviour. Some of the changes reported include increased awareness and mindfulness, reporting and confronting bullying, and responsible online behaviour. With more students benefiting from the knowledge and skills acquired from their training, it has led to more responsible, safer, and respectful online interactions.

Using binomial distribution to find the probability of any student experiencing positive change after being educated on bullying, we made the p value—the chance of any student reporting a positive change in their behaviour—a 50% split between two mutually exclusive events, 'have' or 'have not'. With the total number of trials, n , as 58, and successes, 43, as 50, we found that the likelihood of exactly 43 or more out of 58 students experiencing a positive change was 99.99%, which suggests that receiving this education has been effective in drastically improving the student population.

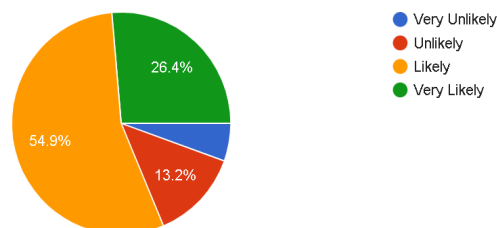


Even though the probability of education improving students' ability to navigate online safely is significantly high at 99.99%, there is still a considerable portion of respondents (25.9%) who did not notice any changes in their online behavior after the education or training. This suggests that the educational intervention may not have been equally effective for everyone. We must consider that there might have been other factors that contributed to this lack of perceived change in behavior to further improve and improve educational efforts in future. The fact that almost $\frac{3}{4}$ of students reported positive changes in their online behavior after education highlights that providing education and training on responsible online behavior is effective in promoting safer, respectful, and more responsible digital practices.

This is also supported based on our survey results, that 74 out of 91 (81.3%) respondents responded that they are likely or very likely to engage in respectful and safe online interactions after being educated about responsible online behaviour and digital citizenship.

How likely are you to engage in respectful and safe online interactions after being educated about responsible online behavior and digital citizenship?

91 responses



However, we must consider that 18.7% of respondents still express reservations about doing so despite receiving education. Overall, the results show that education/training received has positively influenced respondents to engage in safe and respectful online interactions.

Based on the survey results, the sub-hypothesis in this section is proved, as a significant majority of respondents reported positive changes in their online behavior after receiving education on cyberbullying prevention and related topics. With a 74.1% (refer to: page 8, line 2) probability of respondents experiencing a positive change after being educated and 81.3% being likely to engage in respectful interactions online, the hypothesis is further supported.

CONCLUSION

Education on the effects of bullying fosters empathy, respect, and safer interactions amongst youth, reducing bullying incidents. Students educated on digital citizenship have also shown considerable improvements in their behaviour online. Most youths feel safe with school policies in place. However, more than half are unfamiliar with them, so schools should prioritise education over discipline to improve. Significantly less students receive resources on intervening in and mitigating bullying than there should be, thus most feel less confident about intervening and leave cases rampant.

Hence, the analysis of our four sections prove that implementing comprehensive anti-bullying strategies in schools will effectively reduce bullying and cyberbullying incidents among youth.

RECOMMENDATIONS

To reduce bullying incidents, we recommend schools:

1. Shift from disciplinary measures to education. Schools can provide additional assembly talks to raise awareness and understanding.
2. Include role-playing activities in CCE classes that allow students to explore various bullying situations and practice appropriate responses.
3. Establish peer support programs where seniors help juniors. This will foster unity and give a safe forum for students to express their concerns.
4. Make anti-bullying resources publicly available online, enabling individuals to access information and training independently.
5. Teach students conflict resolution skills and strategies for dealing with disagreements in a non-violent and polite manner.

REFERENCES

Singapore Children's Society. (2008, July). *Bullying in Singapore Schools*.
<https://www.childrensociety.org.sg/wp-content/uploads/2022/11/Bullying-in-Singapore-Schools-2008.pdf>

Wei Ling, C. C. (2021, April 1). *Bullying cases in schools*. MOE.
<https://www.moe.gov.sg/news/parliamentary-replies/20210104-bullying-cases-in-schools>